Lab Procedure for Simulink

Play

Setup

1. Hardware Preparation:

    a. Ensure that the QArm Mini is securely attached to the base.

    b. Verify that the manipulator is in the rest position.

    c. Confirm that the QArm Mini is connected to the PC and turn it ON (the light in the switch should be red).

    d. Check and update the latency setting as shown in Figure 1:

        i. Navigate to Device Manager > Ports

        ii. Select the appropriate device - USB Serial Port (COMx) Make a note of the COM port Number.

        iii. Go to Port Settings > Advanced > Latency

        iv. Set the latency to 2 ms

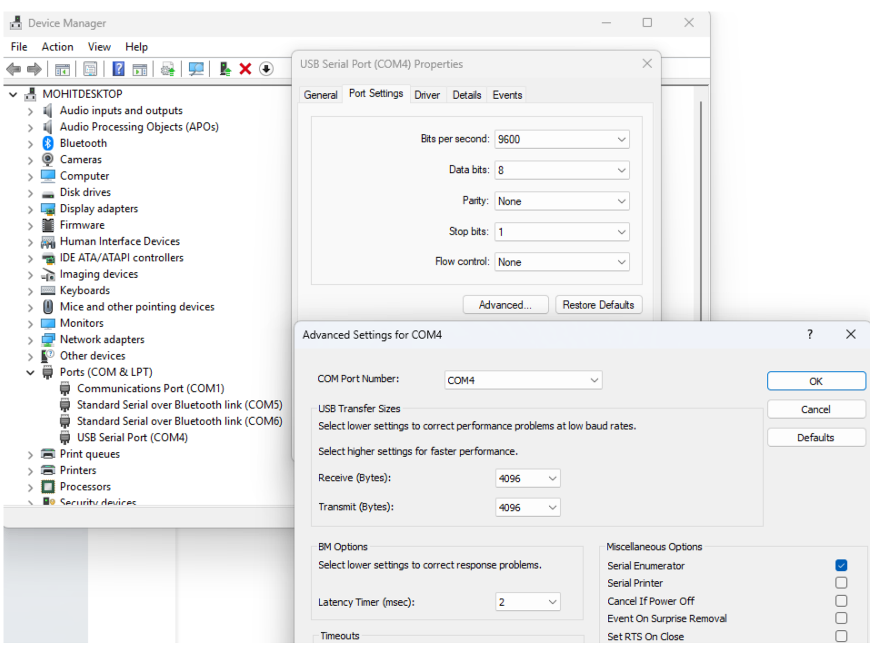

Figure 1. Latency Settings

Workspace Identification

1. Open playslx
    a. In the Hardware tab, open **Hardware Settings -> Solver** and verify:
        i. Solver Type: Fixed
        ii. Solver: ODE2
        iii. Fixed Step Size: 1/30
    b. From the root level's **Application Layer**, double-click QArm Mini subsystem to navigate to the **Interface Layer**, and then the **Hardware Layer**, and double-click on the **HIL Initialize** block.
    c. Update the **Board Identifier** value to match the COM port you noted during setup.
2. Ensure the space around the QArm Mini is clear of any objects. The manipulator will lift itself to home positions as soon as the model starts.
3. Go back to the root level of your model (**Application Layer**). Build and deploy the model using the **Monitor and Tune** button on the Hardware or QUARC tab.
4. The manipulator starts in the home position. You can control each joint individually by adjusting the Joint Selector constant and using the up and down arrow keys on your keyboard.

    1 -> Base Joint

    2 -> Shoulder Joint

    3 -> Elbow Joint

    4 -> Wrist Joint

    Note: You can press **spacebar** at anytime to return the arm to the home position
5. Explore the manipular's limits by moving it to various positions using the keyboard. Observe its range of motion.
6. Adjust the Slider Gain block to control the gripper. You can change the values between 0-to-1, controlling when its fully open, closed or partially open/closed depending on the application.
7. Try to capture all the points in the reachable workspace of the manipulator. Once you have recorded enough positions going through all the joint limits, stop the model.
8. Upon stopping, a graph will be generated, mapping all the end-effector positions reached during the motion of the arm. Does this graph represent the overall workspace of the manipulator well?

    Note: You can use the built-in plot tools in the graph to rotate the 3D render as shown in Figure 2.

9. Repeat steps 3 through 7 if you are not satisfied with the workspace graph generated. When complete, take numerous screenshots of the workspace in the xy, xz and yz planes, as well as an isometric view.

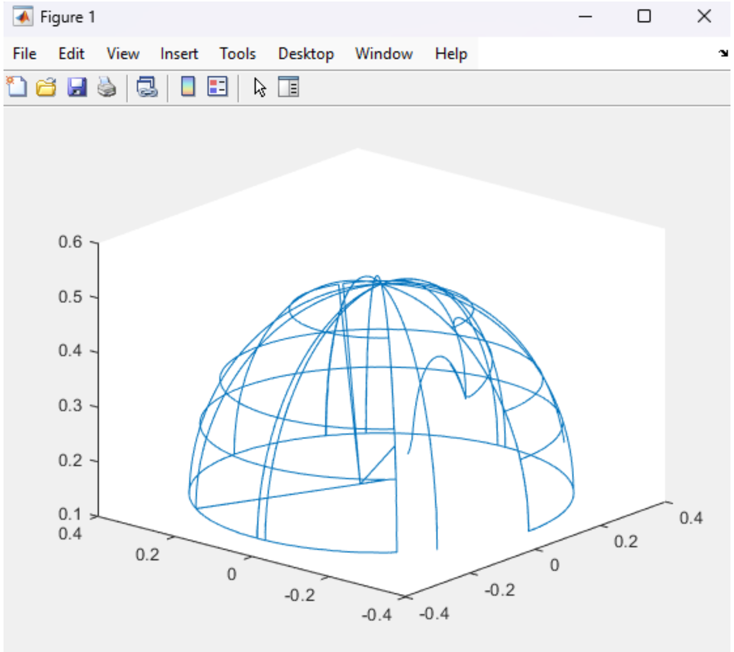

Figure 2. End Effector Positions Mapped

10. Close the graph and the Simulink model.

11. Turn off the arm and gently move it to its resting position (refer User Manual).